

Last three digits before decimal point

RECURRENCE RELATIONS and modulo arithmetic are the topics of the problem ¹ in this note.

Problem

Find the last three digits before the decimal point for the number $(3 + \sqrt{5})^n$. For example, when $n = 5$, $(3 + \sqrt{5})^5 = 3935.73982\dots$, the answer is 935. For $n = 2$, $(3 + \sqrt{5})^2 = 27.4164079\dots$, the answer is 027. The value of n is in the range $2 \leq n \leq 2000000000$.

Looking at the numbers $(3 + \sqrt{5})^n$, we can see that in general they are not integers. Ideally we would like to deal with integers. This sparks the idea of introducing the complement of $(3 + \sqrt{5})$ into the mix, namely $(3 - \sqrt{5})$. Let's look at the binomial expansion ² of $(3 + \sqrt{5})^n$:

$$(3 + \sqrt{5})^n = \sum_{i=0}^n \binom{n}{i} 3^i (\sqrt{5})^{n-i}$$

Compare this to the binomial expansion of $(3 - \sqrt{5})^n$:

$$(3 - \sqrt{5})^n = \sum_{i=0}^n \binom{n}{i} 3^i (-\sqrt{5})^{n-i}$$

When $n - i$ is even, then $(\sqrt{5})^{n-i}$ and $(-\sqrt{5})^{n-i}$ are integers. When $n - i$ is odd, then the binomial terms for $(\sqrt{5})^{n-i}$ and $(-\sqrt{5})^{n-i}$ in the binomial expansions cancel each other out. So it follows that

$$\forall n \in \mathbb{N} : (3 + \sqrt{5})^n + (3 - \sqrt{5})^n \in \mathbb{N}$$

This is encouraging, so we define for all n :

$$a_n = (3 + \sqrt{5})^n$$

$$b_n = (3 - \sqrt{5})^n$$

$$c_n = a_n + b_n$$

¹ Cosmin Negruseri. Code-jam 2008 round 1a: Problem c: Numbers. 2008. URL <https://code.google.com/codejam/contest/32016/dashboard#s=p2>

² Binomial expansion:

$$(a + b)^n = \sum_{i=0}^n \binom{n}{i} a^i b^{n-i}$$

We see that $\forall n \in \mathbb{N} : 0 < b_n < 1$, so $c_n = \lceil a_n \rceil$.

Concentrating on c_n , let's try to find the hundreds digit, the tens digit and the units digit of c_n .

Consider the polynomial:

$$(x - (3 + \sqrt{5}))(x - (3 - \sqrt{5})) = x^2 - 6x + 4$$

It leads to the recurrence relation: $f_n = 6f_{n-1} - 4f_{n-2}$, for which any linear combination of a_n and b_n is a solution³. We set the initial values of f_n such that the linear combination $c_n = a_n + b_n$ is the solution: $f_0 = 2, f_1 = 6$.

Therefore c_n satisfies the recurrence:

$$\begin{aligned} c_n &= 6c_{n-1} - 4c_{n-2} \\ c_0 &= 2 \\ c_1 &= 6 \end{aligned}$$

In theory we could just use this recurrence to compute c_n and then extract the hundreds digit, the tens digit and the units digit. Unfortunately this is not feasible for large n , since c_n grows quickly to very large values. But since we only need the last three digits of the values, we don't need to compute the values completely, computing them modulo 1000 will suffice.

Fortunately according to modulo arithmetic, the recurrence relation for c_n is still valid when doing modulo 1000. Let:

$$d_n \equiv c_n \pmod{1000}$$

and so

$$d_n \equiv 6d_{n-1} - 4d_{n-2} \pmod{1000}$$

Now consider the ordered pairs $(d_n, d_{n+1}), n \in \mathbb{N}$. Because $d_n \in \{0, 1, 2, \dots, 999\}$, there are only 10^6 distinct pairs of (d_n, d_{n+1}) possible. So it must be that there exist two indices $i, j \in \mathbb{N}^+$ such that:

$$(d_i, d_{i+1}) = (d_j, d_{j+1})$$

From the recurrence it follows that:

$$\forall k \in \mathbb{N} : (d_{i+k}, d_{i+k+1}) = (d_{j+k}, d_{j+k+1})$$

The sequence of ordered pairs (d_n, d_{n+1}) is periodic with a period p of at most 10^6 . We can construct a lookup table holding values of d_n from one period p and then compute d_n for large n by going into the lookup table at $n \bmod p$.

The periodic part of the sequence doesn't necessarily start with the first pair in the sequence or with the second or the third etc... There might be a sequence prefix of ordered pairs that don't repeat before it

³ In-depth treatment of recurrence relations can be found in Chapter 10, Ralph P. Grimaldi. *Discrete and Combinatorial Mathematics: An Applied Introduction*. Addison-Wesley, 3rd edition, 1993. ISBN 0201549832

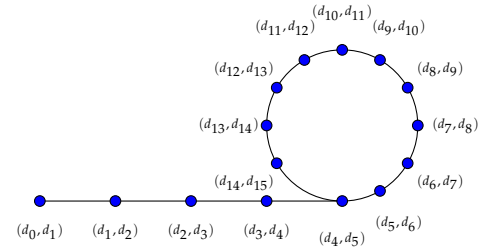


Figure 1.1: Periodic sequence of pairs preceded by a prefix of pairs.

goes into the sequence loop of repeating pairs. We write a function that computes the prefix and the period of the sequence using Floyd's cycle finding algorithm⁴.

Luckily it turns out that in this case the prefix is 2,6,28 and the periodic sequence has period 100.

With the lookup table we can compute d_n for any large n in constant time. d_n gives us the last three digits of c_n . Our goal though was to compute the last three digits before the decimal point of a_n .

We know $c_n = \lceil a_n \rceil$, so $c_n - 1 = \lfloor a_n \rfloor$. This means that to get the digits for a_n , we need to extract them from $d_n - 1$. Listing 1.1 has the complete Haskell implementation.

⁴Floyd's algorithm is described at https://en.wikipedia.org/wiki/Cycle_detection#Tortoise_and_hare. We use the Haskell implementation from https://wiki.haskell.org/Floyd's_cycle-finding_algorithm

Listing 1.1: Haskell code to compute last 3 digits

```

module Last3Digits(
    compute
) where

f :: Int -> Int -> Int
f a b = (6 * b - 4 * a) 'mod' 1000

-- ds is our sequence of d_n
ds = 2 : 6 : zipWith (f) ds (tail ds)
-- this gives us the pairs
dps = zip ds (tail ds)

findCycle :: Eq a => [a] -> ([a],[a])
findCycle xxs = fCycle xxs xxs
    where fCycle (x:xs) (_,y:ys)
        | x == y           = fStart xxs xs
        | otherwise       = fCycle xs ys
    fCycle _ _             = (xxs,[]) -- not cyclic
    fStart (x:xs) (y:ys)
        | x == y           = ([], x:fLength x xs)
        | otherwise       = let (as,bs) = fStart xs ys in (x:as,bs)
    fLength x (y:ys)
        | x == y           = []
        | otherwise       = y:fLength x ys

tps = findCycle dps
-- ps is the prefix, cs the cycle of d_n
(ps, cs) = (map fst (fst tps), map fst (snd tps))

computeAux :: Int -> Int
computeAux n
    | n < (length ps) = ps!!n
    | otherwise       = cs!!((n - (length ps)) 'mod' (length cs))

compute :: Int -> Int
compute n = computeAux n - 1

```

To check our computation we can use this Mathematica function:

```
In[1]:= last3Digits[n_Integer] := Mod[IntegerPart[(3 + Sqrt[5])^n], 1000]
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Bibliography

Ralph P. Grimaldi. *Discrete and Combinatorial Mathematics: An Applied Introduction*. Addison-Wesley, 3rd edition, 1993. ISBN 0201549832.

Cosmin Negruseri. Codejam 2008 round 1a: Problem c: Numbers. 2008. URL <https://code.google.com/codejam/contest/32016/dashboard#s=p2>.